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ARTICLE



Teaching human geography using a couplet game

Chao Ye, Yang Gao, Jie Yu and Yixu He

Faculty of Education, School of Geographic Sciences & Institute of Eco-Chongming, East China Normal University, Shanghai, China

ABSTRACT

Using games to teach is an interesting and worthwhile practice; however, the couplet game has not been described before. In this study, first-year students played a couplet game with their teacher at the end of the course. Each student was asked to summarize and write individual opinions on human geography in one keyword; then, the teacher wrote the corresponding or matching keywords. Analyzing the correspondences and the contrasts between couples of words, the students showed an improved understanding of human geography. The results suggested that the couplet game elicited students' creative and independent thinking. This method can also be applied to other fields.

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Introduction

With the emergence of new media, tools, and methods, teaching and learning are becoming more diversified than ever (Dengler, 2008; Healey & Jenkins, 2000; Ye et al., 2018). The information and communications technology and network have greatly changed the traditional classroom, and an increasing number of teachers use multimedia in their classrooms (Conover & Miller, 2014). Game teaching is another emerging method in this process, which is considered as an important way to increase students' participation (Hwang et al., 2013; Von Wangenheim & Shull, 2009). The electronic games, digital games, serious games, simulation games can be helpful to teaching (Rosser et al., 2007; Sitzmann, 2011; Wouters et al., 2013; Zin et al., 2009). The application of the game teaching method has led to a shift from a traditional teacher-centered learning environment to a student-centered environment, in which students are more active and engaged (Watson et al., 2011).

Game teaching methods are often used in geography. Geography learning usually involves classroom learning and field practice. In class, card games can help students gain a quick understanding of the town (Dove, 2014). The computer simulation model SimCity 2000 can be used as a tool to teach the concept of urban geography (Adams, 1998). In the field practice, the childhood game of capture-the-flag provides a good method for geography students to carry out practice. In the process of capturing the opponents' flags, the students practice their navigation skills and basic ability to assess the terrain (Hupy, 2011). These games play a positive role in geography teaching. The game

teaching method not only enables the students to learn about geography in the classroom, but also improves their field practice ability, achieving the goal of teaching through entertainment.

As each person's personality is different, the choice and design of teaching games are particularly important, and they largely determine the learning effect of students in class (Buckley & Doyle, 2017). If the game is interesting for the students, it will stimulate their enthusiasm for learning (De-Marcos et al., 2016). The introduction of literary works into geography teaching has proved to be an interesting game teaching method, and novels and poems can provide educators with rich information to illustrate the concept of geography (Chamberlain, 1995; Kong & Tay, 1998; Marchetti, 1993). For example, the teaching of poetry takes classical poetry as teaching material, which can be used to review and strengthen the concept of geography (Kirman, 2007; Sekeres & Gregg, 2008; Ye & Wu, 2018).

Despite the widespread use of games in geography teaching, to our knowledge, couplet games have never been used in this context before. Therefore, this study introduces the cultural element of couplets into human geography teaching to fill this gap. Students write a keyword about human geography on the blackboard. The teacher writes couplets. Both the teacher and students are involved in this process. The couplets may not completely resemble a game but the process has many similarities to games. For instance, game teaching can successfully attract students' interests in learning (Jong, 2015; Virvou & Katsionis, 2008; Ye & Ji, 2019), the couplet games also have this characteristic. In addition, the couplet game teaching also has some advantages over the regular game teaching. On the one hand, regular game teaching always sets leaderboards or badges (Hanus & Fox, 2015; Jurgelaitis et al., 2019), the lack of them in couplet games just reduce the pressure brought by regular games. On the other hand, performance – oriented learning process is often present in regular game teaching (Ding et al., 2018; Mekler et al., 2017), but the couplet game emphasizes on students' acquisition of knowledge.

In the teaching process, it is crucial to improve students' learning motivations and promote effective teacher–student interactions (Cordova & Lepper, 1996). Introducing games into class has a brilliant teaching effect (Squire & Barab, 2004). There have been related explorations in integrating cultural elements into geography teaching (Donaldson, 2001). Hence, it is necessary to set up the couplet game. Integrating cultural elements into games, promoting teacher–student interaction, and enhancing students' interests in learning are important methods.

The couplet is one of the most famous traditional cultures in China. Derived from traditional Chinese metrical poem and consisting of two complementary poetic lines, the couplet originated in the tenth century and became increasingly popular during the Ming Dynasty (1368–1644). It is ubiquitous, for example, as a Spring Festival decoration where each line of a couplet is fastened each side of doorway to express good wishes. In the process of playing couplet game, once the first line of a couplet (some words or a brief sentence) is given, the second line of a couplet should be responded quickly and symmetrically, such as coupled words up-down, sky-land, time-space, which must strictly match from the content, meaning and word numbers. When playing the couplet game, the students write down the keywords of human geography on the blackboard and share their thoughts. Then, the teacher gives corresponding words based on the keywords given by the students. This process is so similar to Chinese couplets that it is called

a couplet game. In the couplet game, everything is random and real-time, so the interaction between the teacher and the students is exciting.

The use of couplet games as a teaching method

The study participants were first-year students at the School of Geographical Sciences, a well-known university in China, in the 2019–2020 academic year. The sample included 164 students from different regions of the country. They were majoring in either geographic science or geographic information systems. The compulsory course in which the teaching practice was implemented was the Introduction to Human Geography; the textbook selected for use in this course was “An Introduction to Human Geography: Issues for the 21st Century (3rd ed.)” by Daniels et al. (2008). Before the couplet game course began, the teacher taught the students theoretical courses, which enabled them to have a basic understanding on human geography and master the basic terminology. Moreover, all students had to submit one or two after-class essays.

At the beginning, the teacher chose 13 students who had outstanding performances during the whole term since they might have deeper understanding of geography. They tended to be more active, and could create a lively atmosphere. After that, because of limited class time, 18 volunteers participated in this couplet game, and the rest participated in it through watching the whole process and taking notes.

After the couplet game class, a survey was carried out among the 164 students in the class to explore participants’ assessment on this couplet game and their feedback in details. Although only 1/4 of students had the chance to write down their keywords because of limited time, 152 participants approved of the couplet game, while the rest kept neutral attitude. Two students were dissatisfied with this new game. The specific reasons are described in the case study section. The typical feedback was that the activity was well organized and enabled students to express their ideas more clearly.

The couplet game was first introduced in 2016. The teachers gained experience with each iteration and by 2019, the authors (including a first-year student), began to summarize the highlights and contents of this teaching method using the 2019 class as a case study. The couplet game is different from other games; it does not require a lengthy design, and has a strong operability. The game does not only arouse students’ enthusiasm, but also improves their critical thinking skills. The game can be applied to a specific topic, and it is worthwhile promoting because it is beneficial for teachers to grasp students’ learning focus and thinking.

A case study

Before starting the couplet game, the teacher informed the students that there was only one simple rule: that keywords were limited to two words. The game started with a student writing down the keywords and explaining them. The teacher analyzed them quickly and immediately wrote down the matching keywords, illustrating the reasons for the choices to the students. Finally, these keywords were displayed on the blackboard one by one (see Figure 1). This interactive process could be clearly followed by all students and easily involved them.

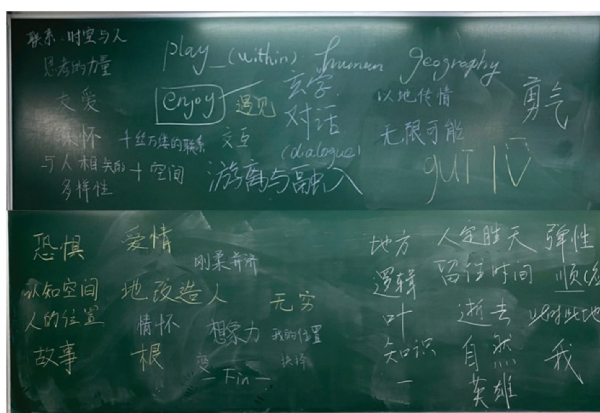


Figure 1. The students' summary words and the teacher's couplet. Source: Authors' photo.

In the following example, the first pair of keywords are interesting and significant. The student wrote “strength of thinking,” and the teacher answered “courage.” Below is their dialogue:

Student: “From where I stand, thinking is the most important ability to learn human geography, and the strength of thinking is the driving force of human geography.”

Teacher: “Don't other subjects also need thinking? Or is their thinking powerless?”

Student: “I regard that thinking is more important in human geography than in other fields.”

Teacher: “I would like to respond her with the word ‘courage’. Her essay title is ‘Life needs the courage to go ahead’. Why do I encourage her to give speech in this class? It is a serendipity. She talked with me before this class. During the talk, I felt that she was depressed because of some difficulties (not in human geography). What I want to say is that the difficulties encountered in the study of human geography are the same as those in any other courses. So, what is most needed is courage. For example, the two great thinkers and geographers, Peter Kropotkin and Bill Bunge whom we talked about last week, could have had a good life if they did the same as the public, but why did they go the opposite way? There are many reasons, but in the end, it is probably the courage that leads them to choose a different life from the ordinary people and scholars, and to make a revolutionary contribution to geography. Courage occupies the foremost place, and what she did and said have proved this. However, the key is whether you put it into practice, dare to face these things, and have the courage to make some changes. Just do it with courage. And this is the reason why I wrote down courage to match her strength of thinking.”

The following are some examples of couplets:

Position – nature

The student thought that humans originally belonged to the nature, then naturally became independent. “We need to figure out what the position of humans in nature is,

whether their position influences their interactions, and how their position affects their interactions.” The teacher replied with the word “nature.” “The position of humans involves their positions in nature and how to position oneself. Nature has two meanings, not only the physical environment, but also the instinct. Your nature is your place.”

Understand space – keep time

The student believed that space belonged to the scope of geographical research, whereas cognition was generated by people. “Human geography focuses on the relationship between humans and spatial geography, that is to say, humans’ cognition of geography, and emphasizes the role of humans in this process.” The teacher responded with “keep time.” “Space and time are a couple of concepts in geography, as Sauer (1974) pointed out time is the fourth dimension of geography. Besides, the verb ‘keep’ matches ‘understand,’ which means action is as important as knowledge. Keeping time and understanding space are significant not only for everyday life, but also for understanding and doing geography.”

Story – hero

Students put forward that all humanistic phenomena were stories with diversity and multiple subjects. “Every story has a space, it is not necessarily a concrete space, but also an abstract space.” The teacher responded with the word “hero.” Combined with the heroic examples mentioned in the previous course, he pointed out, “Many subjects, including human geography, are telling a story, such as how life evolves in natural sciences, and so on. A large number of scientific discoveries have been extended to a story to be told. Thus, curiosity about stories is common to all disciplines. The studies or single things we do, probably end up being a story. Thanks to the hero, the story can be handed down. So, even if we cannot be heroes, we should know at least that the most important thing in this story is the hero.”

Land change – change land

The student had a sense of the migrant workers’ problems mentioned in the previous course. Then, he combined it with environmental determinism and said, “human beings have a certain influence on the geographical environment, but they are also greatly influenced by the surrounding environment.” The teacher accordingly put forward “change land,” the counterpoint of humans’ determination to conquer nature. “On the one hand, the natural environment, humanities, and the social environment are affecting people; on the other hand, people are trying to break many shackles. In connection with the important theme of ancient legends and heroes, their courage and strength to create the world should lead us to ask whether we have this kind of spirit.”

Change – adapt

The student reviewed the previous courses and pointed out that the core concept emphasized by teachers in imparting human geography knowledge was people-oriented. “In the future, when we conduct research on economy, towns and villages,

we should consider the process of the past changes and seek how to make better changes that are conducive to human change.” The teacher further asked, “What is a better change, a change more conducive to change? The change of nature never takes people into account, so how to understand this ‘change?’” The student responded, “Master it, use maps and other means to find the past changes, and find out the reason and law. Although we cannot fully grasp the process and direction of its ‘change,’ we can try our best to use it fully.” Based on this, the teacher gave the word “adapt.” “Perceive such a change, confront this great change, and adapt to the change. Challenge can be interpreted as what pushes us to cope with such a change.”

Fear – elapse

The student thought that human geography originated from a fear of nature, because fear was the emotion that made people study the world they had never been exposed to before. The teacher used the word “elapse” to remind students that everything, including fear, would disappear in the future. “Fear is temporary. Everything, whether it is space or time, will eventually fade away, which is inevitable; all we need to do is to prepare for this. After realizing this, although we will still feel scared, we will have an ordinary heart.”

Root – leaf

A student tried to do a small research project during the course: she had studied a bus route, learned about its stops, and learned about the structure of nearby cities based on the changes in the landscape. In the process of research, she understood how the city developed through historical research, so as to better trace the root of a city, that is, trace the source. The teacher responded with “leaf” and proposed that the most fundamental problem of the research was to trace the root, but after that, knowledge was also needed to support the research, which meant we should understand the original meaning, extend meaning of root and leaf, and learn to use materials and evidence to demonstrate the process.

Imagination – knowledge

A student thought that no matter what the research subject was, from geography to physics, they all required imagination to find more new knowledge and theory. Then, the teacher responded with “knowledge.” Most imagination without knowledge was ultimately empty.

Choose – I

The last student wrote the word “choose.” He said, “First, there is the aforementioned adaptation to change; a person making a choice needs to develop imagination and fosters capacity to make a corresponding choice, so as to adapt to the environment, or to deal with some disasters better. Second, in Western Europe in the middle ages, for example, there were (and still are) people at the top whose choice was to oppress the people below and his own conscience.” The teacher replied, “What is the key to all choices? ‘I.’ All the history, all the outside world, all the choices depend on I in the end.”

The couplet game encourages students to use what they have learned to analyze and condense the knowledge of human geography in their hearts, which cultivates students' dialectical thinking ability and creativity. This can be seen in comparison to the work that students have previously submitted. At the same time, the couplet given by the teacher virtually expands students' thinking. On the one hand, the couplet game has enriched human geography knowledge in students' minds. On the other hand, it has broken the shackles of the inherent framework and created new dimensions. In addition, couplet teaching is also a process of teaching and learning, as an inspiration often sparks some unknown inspirations. Some students' innovative ideas are helpful to inspire teachers to create new research and thinking, forming a new research path. Besides, students' thoughts and feelings about the course help teachers understand the students, thus improving teaching methods and contributing to determining how to develop the course in the future.

At the end of the game, we asked the 164 students to give feedback. Most students claimed that the couplet game mainly focused on the thinking ability, understanding to a more profound level, the expansion of the field of vision, enhanced engagement, human feelings of apprehension, and so on. Two students put forward different opinions, doubting the significance of the game. The selected responses are the most typical from all responses.

The antithetical couplet game represents the immediate responses of teachers and students, and stimulates their thinking in a short time, like a brainstorming. Some students commented:

Students were asked to describe their perception of the world of human geography from various perspectives, and then the teacher responded accordingly to help us establish interdisciplinary thinking and understand human geography from a broader perspective.

The way of thinking I have learned, namely the observation ability, attention ability, thinking ability, and practical ability on human geography, will always accompany me and help me in my future study.

To some extent, the couplet teaching method has trained students' way of thinking and helped them gain a deeper understanding of geography:

This reminds me that geography is not only a subject about space, but also about time. In the communication between time and space, people leave the shadow of time while recording location and understanding space. Such a beautiful geography.

This method facilitates us to realize the diversity and inclusiveness of human geography. Everyone has his own unique understanding, which will guide us to think more deeply from the perspective of human geography in the future study of geography.

This method also plays a role in expanding students' horizons and arousing their interests in learning, as reported by some students:

Although the small game was at the end of the human geography course, I think it may also be the new start point of my human geography study. Human geography is infinite.

This kind of enlightenment will really make people fonder of human geography.

This teaching practice enables students to integrate the charming Chinese couplet culture with human geography. In the feedback, some students also think that this method

deepens self-examination, enhances the understanding of humanistic concerns, and promotes the affection between teachers and students. The course is over, but the impact of the game is far from the end, as students will pursue new thinking and research directions with a renewed interest in the study of human geography.

It is undeniable that there are some problems and difficulties in the implementation of this method. With 164 students in the classroom, it is not realistic to let every student participate in the game in a limited time. Furthermore, the game takes up almost 2/3 of the whole class time, so the teacher lacks time to systematically summarize the words and explain the main points of the couplet. As a result, two students disagree with couplet games. A student said:

The game session took a lot of time. I hope the teacher will give us some time to write, then choose a part to share briefly. The last class is close to the exam, we hope to have a systematic knowledge system.

In addition, due to the particularity of language, some words do not correspond exactly. A student reported, “Not all words can have a corresponding word. It is better to directly carry out a deeper explanation of the first keyword.”

We made a questionnaire specifically to ask students from other grades who had also played this game in the past. Due to the time limitation, 55 former students now in second and third year filled in the questionnaire. Of these responses, 87% approved the couplet game, regarding it positive for enhancing their understanding on human geography and the ability of critical thinking. One of the second years said, “It’s an interesting game which helps cultivate divergent thinking, and spurs us to get a deeper insight of human geography.” Nevertheless, 13% of the students (especially third-year students) do not have enduring memory of the game. This might be related to the passage of time and the increase in the number of courses they learned later.

In sum, antithetical couplet teaching is a new attempt to promote participation and active learning. However, students are typically used to be passive learners; some of them are not adapted to active learning, thus the gains are not obvious. This teaching model is meaningful and reflects the dynamic interactive nature of human geography. The teacher and the students can exchange ideas with each other in the couplet game, and jointly construct and improve the cognitive system for human geography.

Discussion and conclusions

The couplet game breaks the inherent mode of the classroom and builds a platform for free and equal dialogue between teachers and students. During the game, students condensed their cognition of human geography into a keyword and explained it, then the teacher gave another keyword to respond the keyword written by the students and explained it. The whole process was interesting and thoughtful. After the game, all students were invited to comment on it, and the results showed that most students benefited from it. The discussions on the selection and explanation of the keywords deepen students’ observation and thinking, make students understand geography more comprehensively and deeply, and accomplish the goal of strengthening students’ humanistic feelings and responsibilities toward human geography.

The traditional teaching method often emphasizes on knowledge and periodic exams with direct instruction from teachers, but neglects active and deep learning (Beutel, 2010; Jaques, 1992; McCarthy & Anderson, 2000). This pattern solidifies the relationship and role definition between teachers and students. Compared with the traditional teaching method, games including digital games, card games, board games, simulation, and other games play a more and more indispensable role in teaching practice (Gaudart, 1999; Huang & Levinson, 2012; Levine et al., 1980; McDonald & Hannafin, 2003; Watson et al., 2011). However, different games have the different contexts, and produce the different effects. The good game teaching method should pay attention to the diversity of students, enhance interaction between teachers and students, and create something new, which are the characteristics and strengths of couplet game teaching.

In the couplet game, both students and teachers are equal and masters of the game, and they can exchange ideas freely and openly. It is remarkable that the students have to condense their opinions about human geography into one keyword, and the teacher's responses with another matched keyword are instant. This process is exciting, thought-provoking, and creative. In such a one-to-one interaction, students' role changes from a passive receiver to an active explorer. Although couplet game teaching is only employed in the Chinese class currently, it is also worthy to be explored and practiced in other disciplines and countries because coupling or antithesis is common in daily life as well as knowledge.

The couplet game is a creative teaching method, and it is also an important way to emphasize students' dominant role in studying. Given the influence of the traditional teaching mode, some students are used to getting knowledge from the teachers and are unwilling to think actively, meanwhile some of them fail to overcome their shyness and reluctance, which reduces the learning effect of the game. In addition, for those students who participate in the game, due to the limitation of time and their own knowledge reserve, some keywords are difficult for them to fully digest and absorb. Therefore, how to adjust the game so that it can boost students' interests in taking part in the dialogue and interaction with higher efficiency deserves to be pondered in the future.

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No potential conflict of interest was reported by the authors.

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